

1 True or False

- (1) The third layer of the network stack is the transport layer.

Solution: False. The network stack is as follows: L_1 = physical layer, L_2 = datalink layer, L_3 =network, L_4 =transport, L_7 = application.

- (2) Layering in the network stack is an example of the end-to-end principle.

Solution: False. While the end-to-end principle is about what layer functionality should be implemented, layering is a lesson in modularity and abstraction, breaking the functions of the data plane into smaller problems. For example L_3 deals with connecting subnets to each other while L_4 deals with the reliability of flows.

- (3) We implement all the same layers in both the host and routers.

Solution: False. In the hosts we implement layers 1,2,3,4, and 7 but in a router we implement layers 1,2, and 3.

- (4) The operating system (OS) supports logical ports while a router only has physical ports.

Solution: True. A host's OS can create a logical port when using a sockets API in the transport layer while a router cannot.

- (5) Layers 4 and 7 are used for forwarding in switches/ routers. **Solution:** False, there's no need to handle reliability or applications in the network, which means L_4, L_7 are not needed.

2 Protocol Diagram

Refer to Figure 2, which is originally taken from Lecture 4. In this example, Host A sends one packet to host B at time $t = 0$. In this question, we explore how the packet header changes as the packet traverses different layers and protocols of the network. At each time step, fill in the empty blocks to describe which headers are attached to the payload. The packet headers are provided at time $t = 2$ for reference.

Solution:

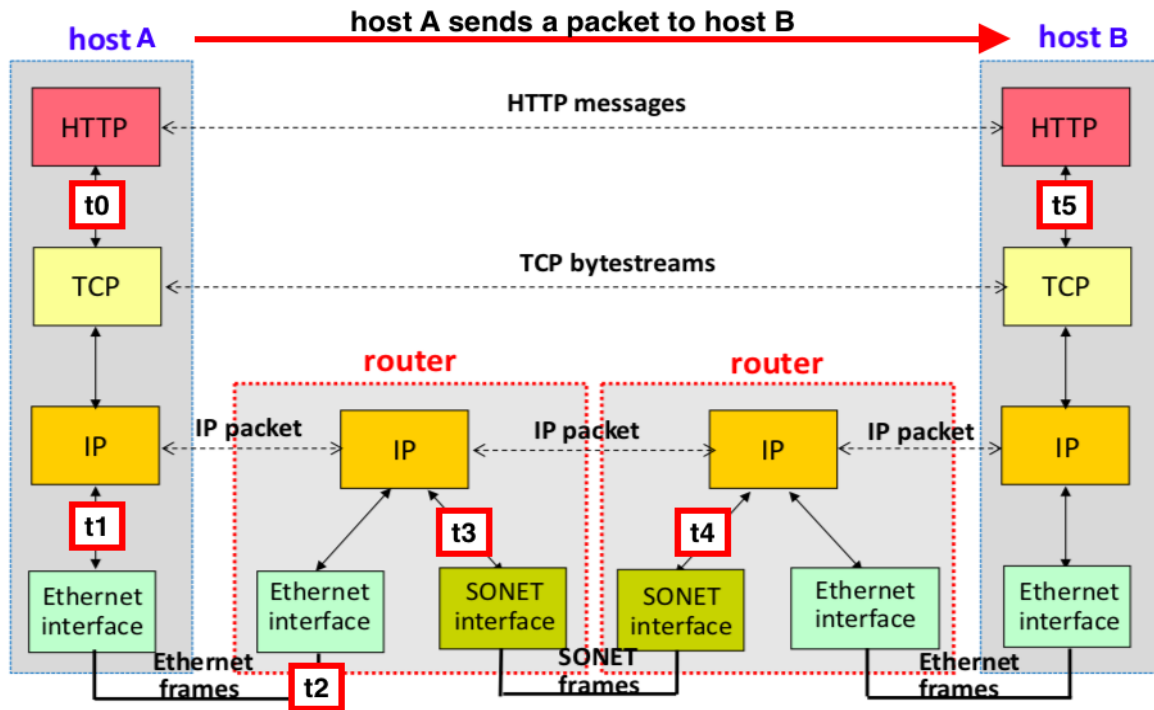


Figure 1: Used for question 2. Host A sends a packet to host B that goes through two routers.

Time = t0					Payload
Time = t1					Payload
Time = t2	L1/L2	L3	L4	L7	Payload
Time = t3					Payload
Time = t4					Payload
Time = t5					Payload

Time = t0				L7	Payload
Time = t1		L3	L4	L7	Payload
Time = t2	L1/L2	L3	L4	L7	Payload
Time = t3		L3	L4	L7	Payload
Time = t4		L3	L4	L7	Payload
Time = t5				L7	Payload

Time = t0				HTTP	Payload
Time = t1		IP	TCP	HTTP	Payload
Time = t2	Ethernet	IP	TCP	HTTP	Payload
Time = t3		IP	TCP	HTTP	Payload
Time = t4		IP	TCP	HTTP	Payload
Time = t5				HTTP	Payload